

Dynamic Gravity (GR): Core Master Equation Sheet

Native Spectral-Projection Framework

Michael Stanislaus Oshetski

ORCID: 0009-0007-3623-7586

May 2026

Purpose and Status

This sheet consolidates the reduced Dynamic Gravity equation framework into one native MR form. The organizing principle is that the validated observable sectors are projections of a single internal spectral system. This version intentionally focuses on the core gravity, Kerr, finite-core, gauge, Higgs, gauge-Higgs closure, gravitational-wave, and validation-test equations. Fermion families, full nuclear structure, atomic structure, and broad particle-sector completion claims are omitted.

The status labels used below are:

D = derived from MR structure, N = normalization/convention, C = constrained by consistency or data,

1 Master Chain

$$\boxed{\Psi \rightarrow \hat{O}_\Psi \rightarrow \{u_n, \mu_n\} \rightarrow P_i[\Psi] \rightarrow \mathcal{O}_i} \quad (1)$$

The universal flow is: master projection field, internal spectral operator, eigenmodes, sector projection, and observable.

$$\boxed{\mathcal{O}_{\text{phys}} = \mathcal{P} \left[\Psi, \hat{O}_\Psi, \{u_n, \mu_n\} \right]} \quad (2)$$

$$\boxed{\Psi \rightarrow \hat{O}_\Psi \rightarrow \{u_n, \mu_n\} \rightarrow \{P_g, P_{\text{rot}}, P_H, P_A, P_X\} \rightarrow \{g_{\mu\nu}, g_{\mu\nu}^{\text{Kerr}}, H, A_\mu, \mathcal{O}_{\text{GW}}\}} \quad (3)$$

2 Internal Hilbert Space, Action, and Spectral System

Equation	Meaning	Tag
$\mathcal{H}_w = L^2(\mathbb{R}, dw)$	(4) Internal projection coordinate as a Hilbert-space coordinate.	D

Equation	Meaning	Tag
$\langle f g\rangle = \int_{\mathbb{R}} dw f^*(w)g(w) \quad (5)$	Inner product defining normalization and projection overlaps.	D
$\hat{O}_{\Psi} \equiv \hat{H}_w = -\frac{d^2}{dw^2} + V_{\Psi}(w) \quad (6)$	Internal Hamiltonian / spectral operator.	D
$\hat{O}_{\Psi} u_n\rangle = \mu_n^2 u_n\rangle \quad (7)$	Internal eigenvalue equation.	D
$\langle u_m u_n\rangle = \delta_{mn}, \quad \sum_n u_n\rangle\langle u_n = \mathbf{1} \quad (8)$	Orthonormality and completeness.	D
$S_{\text{MR}} = \int d^4x d\xi \sqrt{-g_{\text{eff}}} \left[\frac{1}{2} G^{AB} D_A \Psi^\dagger D_B \Psi - V_{\text{MR}}(\Psi, \xi) + \mathcal{L}_{\text{src}} \right] \quad (9)$	Master action over spacetime and internal coordinates.	D
$D_A = \partial_A - iqA_A \quad (10)$	Internal/gauge covariant derivative.	D
$J(x) \propto T_{\mu\nu} u^\mu u^\nu, \quad J(x) \propto T^{00}(x) \text{ in the local rest frame} \quad (11)$	Covariant energy-density source rule.	D
$V_{\Psi}(w) = \frac{1}{2}a_2w^2 + \frac{1}{4}a_4w^4 + \dots \quad (12)$	Local internal projection potential expansion.	D/C
$(a_2, a_4, \dots) = \arg \min E_{\text{vac}} \quad (13)$	Vacuum selection principle replacing arbitrary parameter choice.	C

3 Projection Operators and Sector Map

Operator	Physical sector	Tag
$P_g \sim O_G = V_{\Psi}''(w)$	Gravity as internal stiffness projection.	D

Operator	Physical sector	Tag
P_{rot}	Rotation / frame dragging / Kerr exterior.	D/C
$P_H(w) = w$	Higgs as displacement projection.	D
P_A and S_θ^1	Gauge field and compact phase projection.	D
P_X	Strong-field continuum / ringdown response.	D/C

4 Emergent Geometry and Weak Gravity

Equation	Meaning	Tag
$Q_{\mu\nu} = \langle \partial_\mu \Psi^A \partial_\nu \Psi^B \rangle G_{AB} \quad (14)$	Pre-metric projection correlation kernel.	D
$g_{\mu\nu}^{\text{eff}} = \frac{Q_{\mu\nu}}{ \det Q ^{1/4}} \quad (15)$	Determinant-normalized emergent metric.	D
$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu} + O(q^2) \quad (16)$	Weak-field expansion.	D
$h_{\mu\nu} = q_{\mu\nu} - \frac{1}{4}\eta_{\mu\nu}q, \quad q = \eta^{\alpha\beta}q_{\alpha\beta} \quad (17)$	Trace-subtracted perturbation from determinant normalization.	D
$\rho_P(\mu^2) = \sum_n \langle u_n \hat{V}_g u_0 \rangle ^2 \delta(\mu^2 - \mu_n^2) \quad (18)$	Gravity spectral density.	D
$F(k^2) = \int d\mu^2 \frac{\rho_P(\mu^2)}{k^2 - \mu^2 + i\epsilon} \quad (19)$	Projection self-energy.	D
$F_{\text{eff}}(k^2) = \chi k^2 \tanh \left[\frac{F_{\text{raw}}(k^2)}{\chi k^2} \right], \quad 0 < \chi < 1 \quad (20)$	Saturated self-energy; linear at low response and bounded at high response.	D/C

Equation	Meaning	Tag
$F_{\text{eff}}(0) = 0$ (21)	Massless graviton condition.	C
$D_{\mu\nu\rho\sigma}^{(P)}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - F_{\text{eff}}(k^2) + i\epsilon}$ (22)	Projection-corrected spin-2 propagator.	D
$\frac{F_{\text{eff}}(k^2)}{k^2} \ll 1, \quad k^2 \rightarrow 0$ (23)	GR recovery condition.	C
$G_{\mu\nu}[g_{\text{eff}}] = 8\pi G_P T_{\mu\nu} + \Pi_{\mu\nu}^{(P)}$ (24)	Projection-corrected Einstein equation.	D
$\Pi_{\mu\nu}^{(P)} \rightarrow 0 \Rightarrow G_{\mu\nu} = 8\pi G_P T_{\mu\nu}$ (25)	Low-energy Einstein limit.	D/C
$(k^2 - F(k^2))\Phi(k) = 4\pi G_P \rho(k)$ (26)	MR Poisson equation.	D
$\nabla^2 \Phi = 4\pi G_P \rho, \quad \Phi(r) = -\frac{G_P M}{r}$ (27)	Newtonian limit.	D/C

5 Stability and Quantum/UV Conditions

Equation	Meaning	Tag
$Z^{-1}(k^2) = 1 - \frac{dF_{\text{eff}}}{dk^2} > 0$ (28)	No-ghost residue condition.	C
$\mu_n^2 > 0$ (29)	No-tachyon internal spectrum condition, apart from controlled symmetry-breaking curvature.	C

Equation	Meaning	Tag
	No extra finite- k pole condition.	C
$D^{-1}(k^2) = k^2 - F_{\text{eff}}(k^2) > 0, \quad k^2 > 0$ (30)		
	Stability theorem for saturated self-energy.	D/C
$0 \leq \frac{F_{\text{eff}}(k^2)}{k^2} < \chi < 1 \quad \Rightarrow \quad D^{-1}(k^2) > 0$ (31)		
	Canonical quantization of the MR field.	D
$[\hat{\Psi}(x, \xi), \hat{\Pi}_{\Psi}(y, \xi')] = i\hbar\delta^{(3)}(x-y)\delta(\xi-\xi')$ (32)		
	Mode expansion of the quantized projection field.	D
$\hat{\Psi}(x, \xi) = \sum_n \left[a_n(x)u_n(\xi) + a_n^\dagger(x)u_n^*(\xi) \right]$ (33)		
	Positivity and UV spectral softening for finite self-energy.	C
$\rho_P(\mu^2) \geq 0, \quad \rho_P(\mu^2) \rightarrow 0 \text{ sufficiently fast as } \mu \rightarrow \infty$ (34)		

6 Higgs Projection and Gauge-Higgs Closure

Equation	Meaning	Tag
	Higgs as internal displacement projection.	D
$P_H(w) = w, \quad H(x) = \langle u_0 P_H \Psi(x) \rangle$ (35)		
	Higgs spectral normalization.	D
$Z_H = \sum_n \frac{ \langle u_n w u_0 \rangle ^2}{\mu_n^2}$ (36)		
	Internal projection potential expansion producing symmetry breaking.	D/C
$V_{\Psi}(w) = \frac{1}{2}a_2w^2 + \frac{1}{4}a_4w^4 + \dots, \quad a_2 < 0, \quad a_4 > 0$ (37)		

Equation	Meaning	Tag
$w_0^2 = -\frac{a_2}{a_4} \quad (38)$	Displaced internal vacuum.	D
$v_{\text{phys}} = \sqrt{Z_H} w_0, \quad v_{\text{phys}}^2 = -\frac{Z_H a_2}{a_4} \quad (39)$	Physical scale from projection displacement.	D/C
$\lambda_H = \frac{a_4}{Z_H^2} \quad (40)$	Quartic closure in the current tester convention.	D
$m_H^2 = -\frac{2a_2}{Z_H} = 2\lambda_H v_{\text{phys}}^2 \quad (41)$	Higgs mass as curvature at the displaced projection vacuum.	D/C
$\ell_P = \lim_{k^2 \rightarrow 0} \frac{F_{\text{eff}}(k^2)}{\chi k^2} \quad (42)$	Low-energy gravitational projection leakage amplitude.	D/C
$\mathcal{P}_{HG} = \ell_P ^2 \quad (43)$	Quadratic projection leakage relation.	D/C

7 Gauge and Fine Structure

Equation	Meaning	Tag
$\Psi = \rho e^{iq\theta}, \quad \theta \sim \theta + 2\pi \quad (44)$	Compact phase sector generating gauge covariance.	D
$D_\mu = \partial_\mu - iqA_\mu, \quad A_\mu \rightarrow A_\mu + q^{-1}\partial_\mu\alpha \quad (45)$	Gauge covariance from local compact phase motion.	D
$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad \partial_\mu F^{\mu\nu} = J^\nu \quad (46)$	Maxwell tensor and Maxwell equation.	D

Equation	Meaning	Tag
$\alpha_{\text{MR}}^{-1} = 4\pi Z_A, \quad Z_A = Z_w Z_\theta C_A \quad (47)$	Fine-structure constant from compact gauge normalization.	D/C
$\Lambda_\theta = 5\pi \left(1 - \frac{4}{5\pi^4}\right) \quad (48)$	Current compact winding scale.	C
$Z_\theta = 2 \sum_{m=1}^{\infty} \exp \left[- \left(\frac{m}{\Lambda_\theta} \right)^3 \right] \quad (49)$	Compact winding spectrum; factor 2 is sine/cosine quadrature degeneracy.	D/C
$C_A = \left(1 + \frac{1}{5\pi^4}\right)^{1/4}, \quad N_A = 4+1 = 5 \quad (50)$	Compact gauge-volume correction from four gauge components plus one compact phase constraint.	D/C
$\alpha_{\text{MR}}^{-1} \simeq 137.1877 \quad (51)$	Current fine-structure result from the corrected compact closure.	C

8 Gauge-Higgs Mass Closure

Equation	Meaning	Tag
$Z_A = Z_2 + Z_1 \quad (52)$	Gauge normalization split used in the reduced closure.	D/C
$\sin^2 \theta_W^{\text{MR}} = \frac{1}{1 + \pi + 1/(2\pi)} \simeq 0.232518 \quad (53)$	Compact topology split with Higgs-alignment correction.	C
$Z_2 = Z_A \sin^2 \theta_W, \quad Z_1 = Z_A \cos^2 \theta_W \quad (54)$	Gauge normalization split.	D/C

Equation	Meaning	Tag
	Coupling normalizations.	D
$g^{-2} = Z_2, \quad g'^{-2} = Z_1, \quad e^{-2} = Z_A$ (55)		
	Gauge-Higgs boson masses and exact photon masslessness.	D/C
$m_W^2 = \frac{v^2}{4Z_2}, \quad m_Z^2 = \frac{v^2}{4} (Z_2^{-1} + Z_1^{-1}), \quad m_\gamma^2 = 0$ (56)		
	Compact normalization correction.	C
$C_{EW} = \left(1 + \frac{1}{2\pi^2}\right)^{3/4}, \quad v_{EW} = C_{EW}v_0$ (57)		
	Current first-order mass closure from compact normalization.	C
$m_W^{\text{MR}} \simeq 80.116 \text{ GeV}, \quad m_Z^{\text{MR}} \simeq 91.451 \text{ GeV}$ (58)		

9 MR-Native Kerr and Strong Gravity

Equation	Meaning	Tag
	Rotating MR coherent exterior state.	D/C
$\Psi_{\text{rot}} = R(r)\Theta(\theta)u_0(\xi)e^{-i\omega t + im\phi}$ (59)		
$Q_{t\phi} \propto -\omega m R ^2 \Theta ^2$ (60)	Native origin of frame dragging from projection phase winding.	D
	Separated MR exterior projection equations.	C/O
$\frac{1}{R} \frac{d}{dr} \left[\Delta_r(r) \frac{dR}{dr} \right] + V_r(r) = -\Lambda,$ $\frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left[\sin \theta \frac{d\Theta}{d\theta} \right] + V_\theta(\theta) = \Lambda$ (61)		
$\Sigma_{\text{MR}} = r^2 + a^2 \cos^2 \theta$ (62)	Angular Kerr structure from axisymmetry, reflection symmetry, smooth nonrotating limit, and no exterior hair.	D/C

Equation	Meaning	Tag
$\Delta_{\text{MR}} = r^2 - \frac{2G_P M}{c^2} r + a^2 \quad (63)$	Radial Kerr structure from Schwarzschild limit, flat rotating limit, and no extra exterior charge.	D/C
$a = \frac{J}{Mc} \quad (64)$	Kerr spin parameter in physical units.	D
$ds_{\text{MR}}^2 = - \left(1 - \frac{2G_P M r}{c^2 \Sigma_{\text{MR}}} \right) c^2 dt^2 - \frac{4G_P M a r \sin^2 \theta}{c^3 \Sigma_{\text{MR}}} dt d\phi + \frac{\Sigma_{\text{MR}}}{\Delta_{\text{MR}}} dr^2 + \Sigma_{\text{MR}} d\theta^2 + \left(r^2 + a^2 + \frac{2G_P M a^2 r \sin^2 \theta}{c^2 \Sigma_{\text{MR}}} \right) \sin^2 \theta d\phi^2 \quad (65)$	MR-derived rotating exterior metric; equals Kerr outside the projection core.	D/C
$r_{\pm} = \frac{G_P M}{c^2} \pm \sqrt{\left(\frac{G_P M}{c^2} \right)^2 - a^2} \quad (66)$	Horizon radii of the exterior Kerr branch.	D/C
$\Psi^3 - \Psi = \frac{\rho}{\rho_*}, \quad R_{\text{max}} \sim \mu_{\text{min}}^2, \quad \rho_* \sim \frac{c^2 R_{\text{max}}}{G_P} \quad (67)$	Nonlinear projection saturation, maximum curvature, and density cutoff.	D/C
$r_c = \left(\frac{G_P M}{c^2 R_{\text{max}}} \right)^{1/3}, \quad \frac{r_c}{r_s} \propto M^{-2/3} \quad (68)$	Finite core radius and scaling relative to Schwarzschild radius.	D/C

10 Gravitational Waves and Ringdown

Equation	Meaning	Tag
$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu}, \quad h_{\mu\nu} = \delta \left(\frac{\langle \partial_\mu \Psi \partial_\nu \Psi \rangle}{ \det Q ^{1/4}} \right) \quad (69)$	Gravitational waves are propagating projection-geometry perturbations.	D

Equation	Meaning	Tag
	Native MR gravitational-wave equation.	D/C
$[\square + \Pi_X(\square)] h_{\mu\nu}^{\text{TT}} = 16\pi G_P T_{\mu\nu}^{\text{TT}} \quad (70)$		
	GW propagator with projection-continuum self-energy.	D
$D_{\mu\nu\rho\sigma}^{\text{MR}}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - \Pi_X(k^2) + i\epsilon} \quad (71)$		
	Low-energy propagation is GR-like and luminal.	D/C
$\Pi_X/k^2 \ll 1 \quad \Rightarrow \quad c_{\text{GW}} = c \quad (72)$		
	Kerr response plus internal projection-continuum	D/C
$A(\omega) = A_{\text{Kerr}}(\omega) + A_X(\omega), \quad A_X(\omega) = \int^{\text{response.}} d\mu \frac{\rho_X(\mu)}{\omega - \mu + i\Gamma_X(\mu)} \quad (73)$		

11 Action Variation and Field-Equation Checks

Equation	Meaning	Tag
$\mathcal{L}_\Psi = \frac{1}{2} \partial_\alpha \Psi \partial^\alpha \Psi - V(\Psi) \quad (74)$	Scalar projection Lagrangian.	D
$\frac{\partial \mathcal{L}}{\partial \Psi} - \partial_\alpha \left(\frac{\partial \mathcal{L}}{\partial (\partial_\alpha \Psi)} \right) = 0 \quad (75)$	Euler–Lagrange equation.	D
$\square \Psi + V'(\Psi) = 0 \quad (76)$	Field equation from the scalar projection action.	D
	Gauge-field action with projection normalization.	D
$\mathcal{L}_A = -\frac{1}{4} Z_A F_{\mu\nu} F^{\mu\nu} + J_\mu A^\mu \quad (77)$		
$Z_A \partial_\mu F^{\mu\nu} = J^\nu \quad (78)$	Gauge equation from variation.	D

Equation	Meaning	Tag
$K(k) = k^2 - F(k^2), \quad D(k) = K(k)^{-1}$ (79)	Quadratic action kernel and propagator inverse relation.	D

12 Observable Benchmarks and Test Status

Quantity	Current MR result	Status
Fine structure	$\alpha_{\text{MR}}^{-1} \simeq 137.1877$	Strong first-order; $\sim 0.11\%$
Weak angle	$\sin^2 \theta_W^{\text{MR}} \simeq 0.232518$	Strong first-order
Gauge-Higgs masses	$m_W^{\text{MR}} \simeq 80.116 \text{ GeV}, m_Z^{\text{MR}} \simeq 91.451 \text{ GeV}$	Sub-percent first-order
Weak gravity	GR/Newton recovered for $F/k^2 \ll 1$	Strong
Kerr	MR-native exterior closure with remaining full Ricci symbolic run pending	Strong framework
Finite core	$r_c/r_s \sim 10^{-21}$ for $65M_\odot$ benchmark	Passed toy regularity test
GW propagation	Low-energy propagation recovers GR form	Framework closed

13 Validation Tester Suite

Tester	Purpose	Output File
Symbolic + numerical tester v1	Algebraic closures, gauge-Higgs identities, compact numerical scoreboard	master_relativity_symbolic_ numeric_tester_v1_results.csv
Tensor consistency tester v2.1	Kerr determinant, inverse, limits, Schwarzschild Ricci-flatness	master_relativity_tensor_ consistency_tester_v2_1_results. csv
Action variation tester v3.1	Euler-Lagrange checks, Higgs vacuum, gauge variation, propagator kernel	master_relativity_action_ variation_tester_v3_1_results. csv
Curvature geometry tester v4	Schwarzschild Ricci scalar/tensor, Einstein tensor, Kerr algebraic geometry	master_relativity_curvature_ geometry_tester_v4_results.csv

Tester	Purpose	Output File
Full Kerr Ricci tester v4.3	Full symbolic rotating Kerr Ricci-flatness check	master_relativity_full_kerr_ricci_v4_3_results.csv
Semi-symbolic Kerr Ricci tester v4.4	Faster Kerr Ricci check with $G = M = 1$, $a = 1/2$	master_relativity_semi_symbolic_kerr_ricci_v4_4_results.csv
Finite-core regularity tester v5.1	Saturation, bounded curvature, core scaling, weak-curvature recovery	master_relativity_finite_core_tester_v5_1_results.csv
Weak-field observational tester v6	Mercury precession, light bending, Shapiro delay, weak-field compactness	master_relativity_weak_field_tester_v6_results.csv

14 Parameter Audit

Parameter	Meaning	Tag	Comment
Ψ	Master projection field	D	fundamental
$\hat{O}_\Psi$	Spectral operator	D	internal Hamiltonian
V_Ψ	Projection potential	C/O	selected by E_{vac}
u_n, μ_n	Spectrum	D	eigen-system
$Q_{\mu\nu}$	Metric seed	D	bilinear correlation
$g_{\mu\nu}^{\text{eff}}$	Observable metric	D	determinant normalized
G_P	Newton coupling	C	fixed by low-energy gravity
$F(0) = 0$	Massless graviton	C	required for GR recovery
R_{max}	Maximum curvature	C/O	linked to spectral cutoff
r_c	Finite core radius	D/C	follows from saturation
$\Sigma_{\text{MR}}, \Delta_{\text{MR}}$	Kerr exterior functions	D/C	no-hair/separability assumptions
Z_H	Higgs normalization	D/C	spectral overlap
a_2, a_4	Higgs potential curvature/stiffness	D/C	from internal potential
Z_w, Z_θ, C_A	Fine-structure factors	D/C	compact gauge geometry

Parameter	Meaning	Tag	Comment
$p = 3$	Winding suppression exponent	C/O	compact-volume closure, not uniqueness theorem
Λ_θ	Compact winding scale	C/O	current corrected compact closure
C_{EW}	Gauge-Higgs normalization	C	compact correction
Π_X	GW self-energy	D/C	continuum correction

15 Current Completion Status

- Strongly closed at framework level: master operator, projection metric, weak gravity, Higgs potential, gauge/Maxwell, fine structure, gauge-Higgs first-order masses, finite-core scaling, gravitational-wave propagation.
- Strong but with theorem-level caveats: MR-native Kerr uniqueness, compact winding exponent uniqueness, gauge-Higgs alignment correction.
- Current validation suite: symbolic algebra, tensor identities, action variation, Schwarzschild curvature, finite core regularity, weak-field observations.
- Open frontier: full rotating Kerr Ricci symbolic completion, exact compact-winding theorem, full UV proof beyond EFT-level closure.